

University of Wah

Department of Computer Science



Submit to:

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Course:

Information and Security Lab - CS-251L

Reg No:

UW-24-CS-BS-101

Section:

BS-CS-3rd-B

DATE: _____

Task 09

DATE: _____

Plain Text:

101456A E C D 1 3 1 5 3 4

Step - 1 :-

1 0 0 0 0 1	0 0 0 0 0	1 0 0 0 0	4 0 1 1 0 0	5 0 1 1 0 1
16 17 18 19 20	21 22 23 24 25	26 27 28 29 30	31 32 33 34 35	36 37 38 39 40
10 11 12 13 14	15 16 17 18 19	20 21 22 23 24	25 26 27 28 29	30 31 32 33 34
35 36 37 38 39	40 41 42 43 44	45 46 47 48 49	50 51 52 53 54	55 56 57 58 59

Plaintext after Permutation according to the table that is given in task file (according to table)

0001	0100	1110	0111	1101	1110
0111	1000	0001	1000	1000	1000
1000	0001	1000	0010	1100	

Step - 2 :-

Key = A A B B 0 9 1 8 3 7 3 6 C F D D

A(10) A(10) B(11) B(11) 0 9

1010	1010	1011	1011	0000	1001	0
8	3	7	3	6	C(12)	F
1000	0011	0111	0011	0110	1100	11
D	D					
1101	1101					

GA

DAY

5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	0	1	0	1	0	1	1	1	0	1	1	0	0	0
25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
0	0	0	0	1	0	0	0	0	0	0	1	1	1	1
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55
0	0	1	1	0	1	1	0	1	1	0	1	1	0	0
57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
1	1	0	1	1	1	0	1	1	0	1	1	0	1	1

bit convert

64-bits to 56 bit convert
by table

5	49	41	33	26	17	9	1	58	50	42	34	36	38
6	1	0	0	0	1	1	1	1	0	0	0	0	0
7	1	0	51	43	35	27	19	11	3	60	52	44	36
8	2	59	0	1	1	0	0	1	1	1	0	1	1
9	0	0	0										
10	0	0	0										
11	55	47	39	31	23	15	7	62	54	46	38	30	22
12	1	1	0	0	1	1	1	1	1	1	0	0	0
13	1	61	53	45	37	29	21	13	5	28	20	12	4
14	0	0	1	0	0	1	1	1	1	1	0	1	0
15	0	0	1	0	0	1	1	1	1	1	0	1	0

1000001111000000001100111011
 011100111110000110011111010

By Circular shift:

Left: 10000111100000000011001110111
Right: 1110011111100001100111110100

56 $\rightarrow$ 48 bit by permutation table 2

14	17	11	24	1	5	3	28	15	6	21	10	23	19	12	4
0	0	0	1	1	0	0	1	0	1	0	0	1	1	0	0
16	8	16	7	27	20	13	2	41	52	31	32	47	58	30	40
51	1	0	1	1	0	0	0	0	1	1	1	0	0	1	0
1	1	33	98	44	49	39	56	34	53	46	42	50	36	29	32
		0	1	1	1	1	0	1	0	0	0	1	1	1	0

DATE _____

Plain text:

Left:

Convert Right bit into 48 bit

[illegible]

Taking XOR of Key and R

$$F = R \text{ XOR } K$$

[illegible]

DATE: _____

10 0 1 0 1 0 0 0 0 1 1 0 0 0 0 1
 1 1 0 0 0 0 1 1 0 0 0 0 0 0 1 1 0 0 0 1
 1 1 0 0 0 0 1 1 1 0 0 0 0 0 0 0 1 1 0 1

Making chunks

$C_1 = 100101$ $C_2 = 100001$
 $C_3 = 100001$ $C_4 = 111000$
 $C_5 = 0011000$ $C_6 = 001101$
 $C_7 = 110000$ $C_8 = 111000$

$C_1 = (10)(0101)$ $C_2 = (10)(0001)$
 $C_3 = (10)(0001)$ $C_4 = (11)(1000)$
 $C_5 = (01)(1000)$ $C_6 = (00)(11001)$
 $C_7 = (11)(10000)$ $C_8 = (11)(1000)$

$C_1 = 1000$ $C_2 = 1010$
 $C_3 = 1101$ $C_4 = 1010$
 $C_5 = 0110$ $C_6 = 1111$
 $C_7 = 0011$ $C_8 = 1101$

1000 1010 1101 1010 0110 1111
 0011 1101

P-box permutation

16	7	20	31	29	12	128	17	1	15
1	0	0	0	1	0	1	1	1	0
25	1	5	2	8	24	16	32	24	7
14	13	1	30	0	8	23	0	1	0

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$R_2 = 0010111010101011$
 01010111

$L_2 = R_1$

Swapping whole

0001 0111 1011 1111 0101
1110 1010 0101 0100 1101
0101 0100 1111 1011

64 bits

Cipher text:

→ F083D8E8C2C8
AF4E